Computational and Deep Learning Strategies in Quantitative Finance for Robust Hedging under Market Frictions

MA-500B Thesis Mid Semester Evaluation and Poster Presentation

Comprehensive Technical Report

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Pratham Kailasiya (21323030)
BS-MS Mathematics and Computing
Department of Mathematics, Indian Institute of Technology Roorkee
Thesis Supervisor: Prof. Aditi Gangopadhyay

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1 Introduction

1.1 Problem Statement

The hedging of derivative instruments represents a fundamental challenge in quantitative finance. Traditional Black-Scholes delta hedging black1973pricing assumes continuous trading, zero transaction costs, and complete markets—assumptions that fail in practice. Deep hedging buehler2019deep reformulates this as a stochastic optimal control problem solved via neural network function approximation, naturally incorporating market frictions and risk preferences.

Despite significant progress, several fundamental challenges remain. First, **model uncertainty**: standard deep hedging trains on fixed Heston parameters; distributional shifts during crises degrade performance. Second, **training instability**: the two-stage CVaR→entropic protocol kozyra2019deep exhibits abrupt objective transitions. Third, **regime dependence**: different architectures excel in different market conditions, yet single-model deployment is standard.

We address these challenges with three novel approaches:

1. **W-DRO-T:** Wasserstein DRO regularization for distributional robustness
2. **3SCH:** Mixed-loss homotopy for smooth training transitions
3. **RSE:** Regime-dependent gating over diverse base hedgers

1.2 Market Frictions in Real-World Financial Markets

- **Transaction costs:** Real hedging incurs trading costs (SPY ≈ 3 bps vs NIFTY ≈ 18 bps including STT, stamp duty, and slippage).
- **Discrete rebalancing:** Hedging occurs at discrete intervals (e.g., daily rebalancing with ~ 30 steps for a 30-day option).
- **Model uncertainty:** Stochastic volatility models (e.g., Heston) face parameter instability and calibration errors across market regimes.
- **Incomplete markets:** Stochastic volatility prevents perfect replication of derivatives.
- **Position limits:** Regulatory and risk constraints restrict allowable hedge ratios and exposures.
- **India-specific frictions:** NIFTY/BANKNIFTY options involve STT (0.05%), wider spreads, discrete strikes (50-point intervals), and SEBI position limits.
- **Strategic implication:** High trading frictions favor low-turnover hedging strategies over frequent rebalancing.

1.3 Neural Network Representation of Hedging Policies

The hedging strategy is parameterised as a neural network policy:

$$\delta_k = \delta_{\max} \cdot \tanh(F_\theta(I_{0:k}, \delta_{k-1})), \qquad \delta_{\max} = 1.5, \tag{1}$$

where F_θ is a neural network (LSTM, Transformer, or ensemble) and $I_k = (S_k/S_0, \log(S_k/K), \sqrt{v_k}, \tau_k, \Delta_k^{\mathrm{BS}}) \in \mathbb{R}^5$ is the feature vector. The tanh bounding ensures $|\delta_k| \leq 1.5$, preventing gradient explosions and enforcing economically sensible position limits. The gradient of the delta output:

$$\frac{\partial \delta_k}{\partial h_k} = \delta_{\max}(1 - \tanh^2(W_\delta h_k + b_\delta)) \cdot W_\delta, \tag{2}$$

is bounded by $\delta_{\max}\|W_\delta\|$ and vanishes only in the saturated regime.

1.4 Motivation for Deep Learning in Hedging

Deep hedging buehler2019deep has emerged as the dominant paradigm for learning optimal hedging policies. However, a fundamental vulnerability persists: models trained on paths from a *fixed* stochastic volatility model degrade under distributional shift. LSTM hedgers trained on normal-regime Heston parameters exhibit CVaR_{95} degradation exceeding $4\times$ under COVID-type volatility.

Our audit of baseline models reveals distinct regime-dependent strengths:

- **LSTM:** Best in stable, trending markets (lowest trading volume)
- **Transformer:** Best tail risk during volatile periods (attention captures clustering)
- **Signature models:** Best for path-dependent dynamics (captures path geometry)

2 Background and Literature Review

2.1 Classical Hedging Theory

Black–Scholes delta hedging prescribes $\delta_t^{\text{BS}} = \Phi(d_1)$ where $d_1 = \frac{\log(S/K)+(r+\sigma^2/2)\tau}{\sigma\sqrt{\tau}}$. This assumes geometric Brownian motion, continuous trading, zero transaction costs, and market completeness. Under stochastic volatility (Heston model), the market is incomplete and hedging residuals persist.

2.2 Deep Hedging Framework

buehler2019deep introduced neural-network-parameterised hedging under convex risk measures, demonstrating that learned policies outperform Black–Scholes delta hedging in the presence of transaction costs. kozyra2019deep proposed two-stage curriculum training (CVaR pretraining followed by entropic fine-tuning) and introduced delta bounding via tanh for training stability. horvath2021deep extended the framework to rough volatility models. carbonneau2021deep applied deep hedging to long-term insurance derivatives.

2.3 Attention Mechanisms and Transformer Architectures

vaswani2017attention introduced the Transformer; applications to financial time series include Informer zhou2021informer and SigFormer tong2023sigformer. For hedging, the causal self-attention mechanism captures long-range dependencies in market features that recurrent networks may miss. The causal mask enforces non-anticipativity: step k cannot attend to future steps $k' > k$.

Related DRO work. blanchet2019quantifying established Wasserstein-based DRO theory with tractable gradient-norm duality. esfahani2018data proved finite-sample performance guarantees for data-driven DRO. lutkebohmert2019robust showed robust approaches reduce hedging error during COVID-type volatility.

Related RL work. kolm2019dynamic and cao2021deep explored RL approaches. neagu2025drl compared eight DRL algorithms. haarnoja2018soft introduced SAC; huang2025sac added CVaR constraints.

Ensembles and regime switching. dietterich2000ensemble established ensemble variance reduction theory. hamilton1989new introduced regime-switching models. Our RSE adapts the mixture-of-experts paradigm to hedging.

2.4 Universal Approximation in Neural Networks

LSTMs and Transformers are universal approximators for sequential mappings. Path signatures lyons1998differential provide universal features for path-dependent functionals—the signature uniquely determines the path up to tree-like equivalence.

3 Research Contributions

3.1 Overview of Studied Architectures

Seven architectures evaluated: LSTM, Transformer, W-DRO-T, 3SCH, RSE. All share identical data pipeline, feature engineering, and two-stage baseline protocol. The experimental framework: 10 independent seeds (42–942), bootstrap CIs (10,000 resamples), Holm–Bonferroni corrected paired tests, real-market validation on SPY and NIFTY across four crisis scenarios.

3.2 Novel Approaches Introduced

Table 1: Novel approaches: design rationale.

Model	**Gap**	**Innovation**	**Theory**	**Ref.**
W-DRO-T	Distributional shift	Grad-norm penalty	Wasserstein DRO duality	blanchet2019quantifying
3SCH	Training instability	Mixed-loss homotopy	Multiobjective Optimisation	kozyra2019deep
RSE	Regime dependence	Gated ensemble	Ensemble variance reduction	Gap analysis

3.2.1 Wasserstein DRO Transformer (W-DRO-T)

W-DRO-T optimises worst-case risk over a Wasserstein ball: $\theta^* = \arg\min_\theta \sup_{\mathbb{Q}:W_1(\mathbb{Q},\mathbb{P})\le\varepsilon} \mathbb{E}_{\mathbb{Q}}[\rho(\text{P\&L})]$. Via duality blanchet2019quantifying, this reduces to gradient-norm regularisation. Three-phase training: CVaR pretraining $\to$ DRO-annealed entropic ($\varepsilon : 0 \to 0.1$) $\to$ stress testing. Achieves 46% CVaR improvement during COVID on SPY and 41% Basel III/IV capital savings.

3.2.2 Three-Stage Curriculum Hedger (3SCH)

3SCH inserts a mixed-loss homotopy $\mathcal{L}(\alpha) = \alpha \cdot \text{CVaR}_{0.95} + (1-\alpha) \cdot \rho_\lambda$ with $\alpha : 0.8 \to 0.2$ between CVaR and entropic stages. Justified by Berge's maximum theorem ensuring upper hemicontinuous solution paths. Best NIFTY normal CVaR_{95} of 622.26—optimal for high-cost emerging markets with trading volume of only 0.90.

3.2.3 Regime-Switching Ensemble (RSE)

RSE combines frozen LSTM, Transformer, and Signature hedgers via regime-dependent gating. Regime features (RV, trend, momentum, jump) drive a neural classifier mapping to $K = 4$ regimes; a learnable affinity matrix $A \in \mathbb{R}^{4\times3}$ produces model weights. Best $\text{CVaR}_{95} = 3.109 \pm 0.010$ (-3.29% vs LSTM, $p < 0.0001$, Cohen's $d = -7.41$), lowest CV = 0.30%.

4 Mathematical Framework

4.1 Market Model

4.1.1 Heston Stochastic Volatility Model

We work on $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{P})$ satisfying the usual conditions (right-continuous, complete), supporting a two-dimensional Brownian motion (W^S, W^v) with $\langle W^S, W^v\rangle_t = \rho\, dt$, $\rho \in (-1, 0)$. The

Heston model heston1993closed:

$$dS_t = rS_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S, \qquad S_0 > 0, \tag{3}$$
$$dv_t = \kappa(\theta_v - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad v_0 > 0. \tag{4}$$

Remark 4.1 (Feller Condition)**.** *The variance process remains strictly positive when* $2\kappa\theta_v > \xi^2$. *Our calibration:* $2 \times 1.0 \times 0.04 = 0.08 > 0.04 = \xi^2$.

4.1.2 Model Parameters

$S_0 = K = 100$, $v_0 = 0.04$, $\kappa = 1.0$, $\theta_v = 0.04$, $\xi = 0.2$, $\rho = -0.7$, $r = 0.05$, $T = 30/365$, $N = 30$.

4.2 Discretisation

4.2.1 Euler–Maruyama Scheme

Full-truncation scheme on equidistant grid $\Delta t = T/N$:

$$S_{k+1} = S_k \exp\!\big[(r - \tfrac{1}{2}v_k^+)\Delta t + \sqrt{v_k^+\,\Delta t}\;\epsilon_k^S\big], \tag{5}$$
$$v_{k+1} = v_k + \kappa(\theta_v - v_k)\Delta t + \xi\sqrt{v_k^+\,\Delta t}\;\epsilon_k^v, \tag{6}$$

where $v_k^+ = \max(v_k, 0)$ and $(\epsilon_k^S, \epsilon_k^v) \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{12} = \rho$.

4.3 Deep Hedging Formulation

4.3.1 Hedging P&L Definition

For European call payoff $Z = (S_T - K)^+$ with $N = 30$ hedging steps:

$$\mathrm{P\&L}(\delta) = -Z + \sum_{k=0}^{N-1} \delta_k(S_{t_{k+1}} - S_{t_k}) - c_{\mathrm{tc}} \sum_{k=0}^{N} |\delta_k - \delta_{k-1}| S_{t_k} \tag{7}$$

with $c_{\mathrm{tc}} = 0.001$ and $\delta_{-1} = \delta_N = 0$ (flat entry/exit).

4.3.2 Neural Policy Parameterisation

$\theta^* = \arg\min_\theta \rho(-\mathrm{P\&L}(\delta^\theta))$ with $\delta_k = \delta_{\max} \cdot \tanh(F_\theta(I_{0:k}, \delta_{k-1}))$, $\delta_{\max} = 1.5$.

4.3.3 Feature Engineering

$I_k = (S_k/S_0, \log(S_k/K), \sqrt{v_k}, \tau_k, \Delta_k^{\mathrm{BS}}) \in \mathbb{R}^5$ where $\tau_k = (T - t_k)/T$ is normalised time-to-maturity and Δ_k^{BS} is Black–Scholes delta.

4.4 Risk Measures

4.4.1 Conditional Value at Risk (CVaR)

Definition 4.2 (CVaR)**.** *For loss* $L = -\mathrm{P\&L}$ *at level* $\alpha \in (0, 1)$*:*

$$\mathrm{VaR}_\alpha(L) = \inf\{l \in \mathbb{R} : \mathbb{P}(L \le l) \ge \alpha\}, \tag{8}$$
$$\mathrm{CVaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,du = \mathbb{E}[L \mid L \ge \mathrm{VaR}_\alpha(L)]. \tag{9}$$

CVaR is a coherent risk measure artzner1999coherent: monotonicity, translation invariance, positive homogeneity, and subadditivity.

Proposition 4.3 (Rockafellar–Uryasev Dual rockafellar2000optimization)**.**

$$\mathrm{CVaR}_\alpha(L) = \inf_{\nu\in\mathbb{R}} \Big\{ \nu + \frac{1}{1-\alpha}\,\mathbb{E}\big[(L-\nu)^+\big] \Big\}. \tag{10}$$

Differentiable in θ (a.e.), enabling gradient-based optimisation.

4.4.2 Entropic Risk Measure

Definition 4.4 (Entropic Risk)**.** *With risk aversion $\lambda > 0$:*

$$\rho_\lambda(X) = \frac{1}{\lambda}\log \mathbb{E}[\exp(-\lambda X)]. \tag{11}$$

Dual representation follmer2002convex: $\rho_\lambda(X) = \sup_{\mathbb{Q}\ll\mathbb{P}}\{\mathbb{E}_\mathbb{Q}[-X] - \frac{1}{\lambda}\mathrm{KL}(\mathbb{Q}\|\mathbb{P})\}$.

Remark 4.5 (CVaR vs. Entropic: Qualitative Differences)**.** *CVaR focuses exclusively on the tail (α-quantile and beyond), producing strategies that aggressively protect against worst-case losses but may over-trade. Entropic risk weighs all outcomes exponentially, producing smoother strategies. The abrupt switch between these objectives motivates 3SCH.*

4.4.3 Dual Representations of Risk Measures

Definition 4.6 (Generalised DRO Risk)**.** *Given divergence $D(\mathbb{Q}\|\mathbb{P})$ and radius $\varepsilon > 0$: $\rho_\varepsilon^D(X) = \sup_{\mathbb{Q}:D(\mathbb{Q}\|\mathbb{P})\le\varepsilon}\mathbb{E}_\mathbb{Q}[-X]$. For $D = \mathrm{KL}$: entropic risk ($\varepsilon = 1/\lambda$). For $D = W_1$: Wasserstein DRO.*

Remark 4.7 (Comparison of Ambiguity Sets)**.** ***KL ball**: requires $\mathbb{Q} \ll \mathbb{P}$, adversary can only* reweight *existing scenarios.* ***Wasserstein ball**: does* not *require absolute continuity, adversary can* move *mass to new regions. For crisis hedging, Wasserstein is more appropriate because distributional shifts involve* new *volatility regimes (COVID $\sigma = 65\%$) not seen during training ($\sigma \approx 20\%$).*

Definition 4.8 (1-Wasserstein Distance)**.** *For probability measures μ, ν on $(\mathbb{R}^d, \|\cdot\|_2)$:*

$$W_1(\mu,\nu) = \inf_{\gamma\in\Gamma(\mu,\nu)} \int \|x-y\|_2\, d\gamma(x,y). \tag{12}$$

Kantorovich–Rubinstein duality: $W_1(\mu,\nu) = \sup_{\|f\|_{\mathrm{Lip}}\le 1}\{\int f\,d\mu - \int f\,d\nu\}$.

5 Base and Novel Hedging Architectures

5.1 Base Hedging Architectures

5.1.1 LSTM Hedger Architecture

5.1.2 Transformer Hedger Architecture

5.1.3 Signature-Based Hedger

Path signatures lyons1998differential: $\mathrm{Sig}(X)_{[s,t]} = (1, \int_s^t dX_u, \int_s^t\int_s^u dX_v dX_u, \ldots)$ truncated at depth m. Captures order of events, oscillation patterns, roughness, and non-Markovian effects. Running statistics (mean μ_t, std σ_t, max/min) of signature features are fed to an MLP producing path-dependent hedge ratios. Used as frozen base model within RSE.

5.2 Regime-Switching Ensemble (RSE)

5.2.1 Ensemble Variance Reduction Theory

Classical ensemble theory dietterich2000ensemble shows that combining M diverse models reduces variance. For equal-weight averaging:

$$\mathrm{Var}(\bar{f}) = \frac{\bar{\sigma}^2}{M} + \frac{M-1}{M}\,\bar{\rho}\,\bar{\sigma}^2 \tag{13}$$

where $\bar{\rho}$ is average pairwise correlation and $\bar{\sigma}^2$ is average individual variance. When $\bar{\rho} < 1$, the ensemble variance is strictly less than the average individual variance, with improvement proportional to model diversity $(1-\bar{\rho})$.

5.2.2 RSE Architecture Overview

RSE consists of four components operating in sequence:

Definition 5.1 (Regime-Switching Ensemble)**.** *The ensemble delta at time step k is:*

$$\delta_k^{\mathrm{RSE}} = \sum_{m=1}^{M} w_m(r_k) \cdot \delta_k^{(m)} \tag{14}$$

where $\delta_k^{(m)}$ are base model outputs, r_k is the detected regime vector, and $w_m(r_k) = \sum_{j=1}^{K} p(r_k = j) \cdot [\mathrm{softmax}(A)]_{j,m}$ are regime-dependent model weights derived from the gating network.

5.2.3 Regime Feature Extraction

We extract six regime-relevant features from price paths:

Definition 5.2 (Regime Features)**.** *At time step k, the regime feature vector is:*

$$r_k = \left(\mathrm{RV}_k^{(5)},\ \mathrm{RV}_k^{(10)},\ \mathrm{RV}_k^{(20)},\ \mathrm{Trend}_k,\ \mathrm{Mom}_k,\ \mathrm{Jump}_k\right) \tag{15}$$

Realized Volatility at windows $w \in \{5, 10, 20\}$:

$$\mathrm{RV}_k^{(w)} = \sqrt{\frac{1}{w}\sum_{i=k-w+1}^{k} (\log S_i - \log S_{i-1})^2} \tag{16}$$

Trend Indicator (SMA crossover):

$$\mathrm{Trend}_k = \frac{\mathrm{SMA}_5(S_k) - \mathrm{SMA}_{20}(S_k)}{\mathrm{SMA}_{20}(S_k)} \tag{17}$$

Momentum:

$$\mathrm{Mom}_k = \frac{S_k - S_{k-5}}{S_{k-5}} \tag{18}$$

Jump Indicator (Barndorff-Nielsen ratio):

$$\mathrm{Jump}_k = \frac{\mathrm{RV}_k^{(20)}}{\mathrm{BV}_k^{(20)}} \tag{19}$$

where $\mathrm{BV}_k^{(w)} = \frac{\pi}{2}\sum_i |r_i| \cdot |r_{i+1}|$ is bipower variation, and $\mathrm{Jump} > 1$ indicates jump activity.

5.2.4 Regime Classifier

An MLP maps features to $K = 4$ regime probabilities:

$$p(r_k = j \mid I_{0:k}) = \text{softmax}\big(f_\phi(r_k)\big)_j \tag{20}$$

where f_ϕ is a 3-layer network: $\mathbb{R}^6 \to \mathbb{R}^{64} \to \mathbb{R}^{64} \to \mathbb{R}^4$ with ReLU activations and batch normalization.

The four regimes correspond approximately to: (1) Low volatility / calm, (2) High volatility / stressed, (3) Trending, (4) Mean-reverting / choppy.

5.2.5 Regime-Dependent Gating Network

A learnable affinity matrix $A \in \mathbb{R}^{K \times M}$ maps regimes to model weights:

$$w_m(k) = \sum_{j=1}^{K} p(r_k = j) \cdot \underbrace{[\text{softmax}(A/\tau)]_{j,m}}_{\text{regime-to-model affinity}} \tag{21}$$

where $\tau > 0$ is the softmax temperature (set to $\tau = 1$ in our experiments). The softmax over columns ensures that weights sum to 1 across models for each regime. The affinity matrix A is the key learnable component: it encodes which base model is best for each regime.

Remark 5.3 (Softmax Temperature and Gating Sharpness)**.** *The temperature τ controls the sharpness of the gating distribution:*

- *As $\tau \to 0^+$, $\text{softmax}(A/\tau) \to$ one-hot (hard switching between models).*
- *As $\tau \to \infty$, $\text{softmax}(A/\tau) \to \mathbf{1}/M$ (uniform equal weighting).*

*With $\tau = 1$, the learned affinity matrix (Table **??**) shows weights ranging from 0.18 to 0.54, indicating soft blending rather than hard switching. This soft gating is preferable for hedging because it avoids discontinuous strategy changes at regime boundaries, which would generate unnecessary transaction costs.*

5.2.6 Training Protocol

RSE training follows a three-step procedure:

Total compute: $3 \times 30 + 20 + 30 = 140$ model-epochs, equivalent to $\sim$80 single-model epochs accounting for the lightweight gating network.

5.2.7 Regime Classification Model Validation

Validated via: (1) high-vol regime probability correlates with RV ($\rho > 0.7$); (2) trend probability correlates with SMA crossover ($\rho > 0.5$); (3) temporally smooth transitions.

Algorithm 1 RSE Training Protocol

Require: Data $\mathcal{D}$, base model architectures

Step 1: Pre-train base models (30 epochs each)
for $m \in \{\text{LSTM, Trans., Sig.}\}$ **do**
    Train F_{θ_m} with $\text{CVaR}_{0.95}$ loss
    Freeze: $\theta_m \leftarrow \text{const}$
end for

Step 2: Train gating (CVaR) (20 epochs)
Trainable params: ϕ (classifier), A (affinity)
for epoch $= 1$ to 20 **do**
    for batch $(I, S, Z) \sim \mathcal{D}$ **do**
        $\delta^{\text{RSE}} \leftarrow \sum_m w_m(r) \cdot F_{\theta_m}(I)$ Eq. (14)
        $\mathcal{L} \leftarrow \text{CVaR}_{0.95}(-\text{P\&L})$
        Update ϕ, A via Adam ($\text{lr} = 10^{-3}$)
    end for
end for

Step 3: Fine-tune gating (Entropic) (30 epochs)
for epoch $= 1$ to 30 **do**
    for batch $(I, S, Z) \sim \mathcal{D}$ **do**
        $\mathcal{L} \leftarrow \rho_\lambda(-\text{P\&L})$
        Update ϕ, A via Adam ($\text{lr} = 10^{-4}$)
    end for
end for
return ϕ^*, A^*

5.2.8 Regime–Model Affinity Matrix

Table 2: Learned affinity (typical run, after softmax).

Regime	LSTM	Transformer	Signature
Low vol (calm)	**0.52**	0.28	0.20
High vol (stress)	0.18	**0.54**	0.28
Trending	0.31	**0.41**	0.28
Mean-reverting	**0.45**	0.22	0.33

LSTM dominates calm/mean-reverting regimes; Transformer dominates high-vol/trending; Signature provides consistent 20–33% weight (path-dependent regularization). This interpretability supports model risk governance.

5.2.9 Training Protocol for RSE

Algorithm 2 RSE Training Protocol

1: **Step 1:** Pre-train base models (30 ep each, $\text{CVaR}_{0.95}$ loss), freeze θ_m
2: **Step 2:** Train gating (ϕ, A) with $\text{CVaR}_{0.95}$ (20 ep, lr 10^{-3})
3: **Step 3:** Fine-tune gating with ρ_λ (30 ep, lr 10^{-4})
4: **return** ϕ^*, A^*

Total: 140 model-epochs ($\sim$80 single-model equivalent). Training: $\sim$924s/seed.

5.2.10 Complexity Analysis

Time: $O(M \cdot C_{\max} + C_{\text{gate}})$ where $C_{\max} = O(N^2 d^2)$. **Space:** $O(\sum_m |\theta_m| + |\phi| + KM)$. Affinity: $KM = 12$ params. **Inference:** < 10ms (parallel GPU).

5.2.11 Model Risk Governance for RSE

We structure RSE governance along SR 11-7 (OCC) and SS1/23 (PRA) pillars.

1. Interpretability. RSE offers *unique* interpretability among the novel approaches through three transparent mechanisms:

- *Regime probabilities* $p(r_k = j)$ at each time step reveal the classifier's market-state assessment — inspectable in real time.
- *Model weights* $w_m(r_k)$ show exactly how much each base hedger contributes — full attribution.
- *Affinity matrix* $A \in \mathbb{R}^{4\times 3}$ provides a static summary of learned regime–model associations (Table **??**), auditable without running the model.

This three-level interpretability stack (time-varying regime $\rightarrow$ time-varying weights $\rightarrow$ static affinity) satisfies explainability requirements that opaque single-model approaches cannot.

2. Modular Validation. Base models and gating network can be validated *independently*:

- Each base model (LSTM, Transformer, Signature) validated against its own CVaR_{95} benchmark.
- Regime classifier validated via correlation of high-vol regime probability with realised volatility ($\rho > 0.7$).
- Gating network validated via end-to-end ensemble CVaR_{95} improvement over equal-weight baseline.

3. Versioning and Independent Updates. Base models and gating network are separately versioned. A base model can be retrained without re-optimising the gating network (and vice versa), enabling incremental updates with minimal disruption.

4. Graceful Rollback. Three fallback levels: (i) set $w_m = 1/M$ for equal-weight ensemble; (ii) select the single best base model for the current regime; (iii) revert to LSTM baseline entirely. All fallbacks require zero retraining.

5.2.12 Hedge Accounting and Regulatory Disclosure

RSE qualifies for hedge accounting under IAS 39/IFRS 9/Ind-AS 109 with dollar offset 0.95 and $R^2 = 0.88$. Under IFRS 7 disclosure requirements, firms should note: (i) the ensemble nature and regime-switching mechanism; (ii) the number and type of base models; (iii) the regime classifier's feature inputs; (iv) the affinity matrix interpretation. Under Basel III/IV FRTB, RSE's lower P&L volatility and stability (CV = 0.30%) support IMA backtesting compliance.

Table 3: Ablation: number of base models

Ensemble	**CVaR$_{95}$**	**CV%**	**Δ vs. Full**
LSTM only	3.215	0.55	+3.4%
LSTM + Trans.	3.142	0.38	+1.1%
Full (3 models)	**3.109**	**0.30**	—

Table 4: Ablation: number of regimes K

K	**CVaR$_{95}$**	**Note**
2	3.128	Under-specified
3	3.115	Competitive
4 (default)	**3.109**	Best
6	3.112	Diminishing returns

5.2.13 Economic Impact

For a \$10B derivatives desk, RSE offers:

- \$135M annual capital savings vs. LSTM
- Transaction costs: \$32M (US) vs. \$18M (LSTM)—incremental cost \$14M
- Net benefit: ∼\$121M annually from capital savings alone

5.2.14 Ablation: Number of Base Models

Each additional base model provides diminishing but meaningful CVaR improvement (Table 3). The Signature model contributes +1.1% additional improvement beyond the two-model ensemble.

5.2.15 Ablation: Number of Regimes

$K = 4$ regimes is optimal (Table 4); more regimes provide negligible improvement while increasing the learnable parameter count.

5.3 Three-Stage Curriculum Hedger (3SCH)

5.3.1 Motivation: Instability in Two-Stage Training

The two-stage protocol kozyra2019deep (CVaR pretraining → entropic fine-tuning) creates a discontinuous change in the optimization landscape. The optimal strategy under CVaR (worst 5%) differs qualitatively from the entropic optimum (exponential weighting). This causes: (1) gradient instability at the stage boundary; (2) forgetting of tail risk improvements; (3) convergence to suboptimal local minima.

5.3.2 Curriculum Learning Perspective

Following bengio2009curriculum, CVaR (tail-focused) is the "easier" objective that constrains the model before the "harder" global entropic objective. 3SCH formalises this transition as a homotopy with provable regularity.

5.3.3 Multi-Objective Optimization Interpretation

The mixed loss $\alpha\,\mathrm{CVaR} + (1-\alpha)\,\rho_\lambda$ is a scalarisation of a bi-objective problem. Homotopy continuation—smoothly varying a parameter to transform one problem into another—is well established in nonlinear programming.

5.3.4 LSTM Base Architecture

3SCH uses the identical LSTM hedger (2 layers, $h = 50$, $\sim$54K params) as its base. The contribution is *purely in the training protocol*—any improvement is attributable solely to the curriculum, not model capacity.

5.3.5 Standard Two-Stage Training Protocol

- **Stage 1 (50 ep):** $\mathcal{L}_1 = \mathrm{CVaR}_{0.95}(-\Pi)$, lr $= 10^{-3}$, patience 15
- **Stage 2 (30 ep):** $\mathcal{L}_2 = \rho_\lambda(-\Pi) + \gamma \cdot \mathrm{TradingPenalty} + \nu \cdot \mathrm{BandPenalty}$, lr $= 10^{-4}$, patience 10

5.3.6 Mixed Loss Formulation

Definition 5.4 (Mixed CVaR-Entropic Loss)**.** *For annealing parameter $\alpha \in [0,1]$:*

$$\mathcal{L}_{\mathrm{mixed}}(\alpha;\theta) = \alpha \cdot \mathrm{CVaR}_{0.95}(-\mathrm{P\&L}^\theta) + (1-\alpha) \cdot \rho_\lambda(-\mathrm{P\&L}^\theta) \tag{22}$$

This is a convex combination of two convex risk measures, hence itself a convex risk measure for each fixed α.

5.3.7 Annealing Schedule

$$\alpha(t) = 0.8 - 0.6 \cdot \frac{t}{T_2 - 1}, \quad t = 0, 1, \ldots, T_2 - 1 \tag{23}$$

Starting at $\alpha = 0.8$ (not 1.0) ensures 20% entropic gradient signal from the first epoch. Ending at $\alpha = 0.2$ retains CVaR regulariser before Stage 3 takes over.

5.3.8 Three-Stage Training Protocol

Algorithm 3 Three-Stage Curriculum Hedger (3SCH)

```
Stage 1 (T_1 = 50 ep): L = CVaR_0.95, lr = 10^-3, patience 15
Store reference deltas δ^ref ← F_θ(I)
Stage 2 (T_2 = 20 ep): α anneals 0.8 → 0.2, lr = 5 × 10^-4
for epoch = 1 to T_2 do
    α ← 0.8 − 0.6(t − 1)/(T_2 − 1)
    L ← α · CVaR_0.95 + (1 − α) · ρ_λ
end for
Stage 3 (T_3 = 30 ep): L = ρ_λ + γ|Δδ|, lr = 10^-4, patience 10
return θ*
```

Total: 100 epochs (vs 80 for two-stage). The additional 20 are for the homotopy transition.

5.3.9 Trading Penalty Formulation

Definition 5.5 (Trading Penalty)**.** TradingPenalty(δ) $= \gamma \sum_{k=0}^{N-1} |\delta_k - \delta_{k-1}|$ *with* $\gamma = 10^{-3}$.

5.3.10 No-Trade Band Penalty

Definition 5.6 (No-Trade Band Penalty)**.** BandPenalty(δ) $= \nu \sum_k \max(0, |\delta_k - \delta_k^{ref}| - \epsilon)$ *with* $\nu = 10^8$, $\epsilon = 0.15$.

Prevents catastrophic forgetting: discourages deviation from CVaR-optimal strategy from Stage 1.

5.3.11 Complexity Analysis

Time: $O(T_{\text{total}} \cdot |\mathcal{D}| \cdot C_{\text{model}})$; mixed loss adds negligible overhead. **Space:** $O(|\theta| + |\mathcal{D}| \cdot N)$ for parameters and reference deltas. Training: $\sim$295s/seed.

5.3.12 Capital Requirements

Under Basel III/IV FRTB, 3SCH achieves capital savings of 22.1% vs. LSTM in US markets (capital proxy: \$6.32M vs. \$8.11M per \$100M notional). While lower than W-DRO-T's 41%, the capital savings come with substantially lower operational costs.

5.3.13 Hedge Accounting

3SCH qualifies for hedge accounting under IAS 39/IFRS 9/Ind-AS 109 with dollar offset 0.95 and $R^2 = 0.88$. Under IFRS 9's principles-based effectiveness testing, 3SCH's lower earnings volatility (Std P&L = 3.70 in SPY normal) supports stable reported earnings from hedging activities.

5.3.14 Model Risk Governance for 3SCH

We structure 3SCH governance along the pillars of SR 11-7 (OCC) and SS1/23 (PRA).

1. Interpretability. 3SCH uses the *identical* LSTM architecture as the baseline—the only change is the training schedule. This is a significant governance advantage: the model itself requires no additional interpretability tooling beyond what exists for the LSTM baseline. The curriculum schedule ($\alpha_{\text{start}}, \alpha_{\text{end}}, T_1, T_2, T_3$) is fully documented, deterministic, and auditable. Each stage's loss trajectory can be inspected independently.

2. Stage-by-Stage Validation.

- *Stage 1:* Verify CVaR_{95} convergence and early-stopping activation.
- *Stage 2:* Verify $\mathcal{L}_{\text{mix}}$ decreases monotonically as α anneals; inspect gradient-norm stability.
- *Stage 3:* Verify entropic loss converges; inspect trading penalty magnitude.
- *End-to-end:* Compare final CVaR_{95} and Std P&L against LSTM baseline; compute bootstrap CIs.

3. Retraining and Rollback. 3SCH inherits the LSTM retraining cadence (weekly recommended). Rollback is trivial: setting $T_2 = 0$ recovers the exact two-stage baseline, requiring no code changes. This "zero-cost rollback" property makes 3SCH uniquely low-risk for production deployment.

4. Regulatory Disclosure. Under IFRS 9/Ind-AS 109, firms should disclose: (i) the three-stage training protocol and annealing schedule; (ii) that the model architecture is identical to the LSTM baseline; (iii) stage-by-stage validation metrics; (iv) the theoretical justification for the homotopy

(Berge's theorem, Corollary **??**). For Indian markets, SEBI documentation requirements are satisfied by the reproducibility appendix.

5.4 Wasserstein DRO Transformer (W-DRO-T)

5.4.1 Distributional Shift in Deep Hedging

Models trained on $\sigma \approx 20\%$ (normal Heston) exhibit CVaR degradation $> 4\times$ under COVID-type $\sigma = 65\%$. This distributional shift is the core vulnerability of standard deep hedging.

5.4.2 Distributionally Robust Optimization Framework

Definition 5.7 (Wasserstein DRO Hedging)**.**

$$\theta^* = \arg\min_{\theta} \sup_{\mathbb{Q}:\, W_p(\mathbb{Q},\mathbb{P}) \le \varepsilon} \mathbb{E}_{\mathbb{Q}}\big[\rho(\mathrm{P\&L}(\delta^{\theta}))\big] \tag{24}$$

where $\varepsilon > 0$ controls the ambiguity set $\mathcal{P}^W(\mathbb{P};\varepsilon) = \{\mathbb{Q} : W_p(\mathbb{Q},\mathbb{P}) \le \varepsilon\}$.

Remark 5.8 (Entropic–Utility Duality)**.** *ρ_λ is the certainty equivalent of the exponential utility $u(x) = -\exp(-\lambda x)$; equivalently, it is the unique solution to $u(\rho_\lambda(X)) = \mathbb{E}[u(X)]$. It is smooth, strictly convex in distribution, and admits a dual representation as an optimised certainty equivalent follmer2002convex:*

$$\rho_\lambda(X) = \sup_{\mathbb{Q} \ll \mathbb{P}} \Big\{ \mathbb{E}_{\mathbb{Q}}[-X] - \frac{1}{\lambda}\,\mathrm{KL}(\mathbb{Q}\|\mathbb{P}) \Big\}, \tag{25}$$

where KL *denotes Kullback–Leibler divergence. This connects entropic risk to information-theoretic robustness: minimising ρ_λ is equivalent to worst-case expected loss over a KL ball, foreshadowing the Wasserstein analogue in Section* **??**.

5.4.3 From KL to Wasserstein: A Unified Robustness View

We now place the entropic risk dual (25) and the Wasserstein DRO objective into a unified framework, clarifying why we prefer the latter for deep hedging.

Definition 5.9 (Generalised Distributionally Robust Risk)**.** *Given a divergence functional $D(\mathbb{Q}\|\mathbb{P})$ and radius $\varepsilon > 0$, the DRO risk is:*

$$\rho^D_\varepsilon(X) = \sup_{\mathbb{Q}:\, D(\mathbb{Q}\|\mathbb{P}) \le \varepsilon} \mathbb{E}_{\mathbb{Q}}[-X]. \tag{26}$$

For $D = \mathrm{KL}$, one recovers entropic risk (25) *(with $\varepsilon = 1/\lambda$). For $D = W_1$, one obtains the Wasserstein DRO objective* (27).

Remark 5.10 (Comparison of Ambiguity Sets)**.** *The choice of D fundamentally affects the nature of robustness:*

1. ***KL ball*** $\{Q : \mathrm{KL}(\mathbb{Q}\|\mathbb{P}) \le \varepsilon\}$*: requires $\mathbb{Q} \ll \mathbb{P}$ (absolute continuity), so the adversary can only* reweight *existing scenarios. Cannot generate new tail events absent from training data.*
2. ***Wasserstein ball*** $\{\mathbb{Q} : W_1(\mathbb{Q},\mathbb{P}) \le \varepsilon\}$*: does* not *require absolute continuity, so the adversary can* move *probability mass to new regions of the input space. This enables robustness to support misspecification — e.g., fat tails absent from Heston training data.*

For hedging during crises, the Wasserstein formulation is more appropriate because distributional shifts involve new *volatility regimes (e.g., COVID $\sigma = 65\%$) not seen during training ($\sigma \approx 20\%$), rather than mere reweighting of existing scenarios. The dual gradient-norm penalty is also more stable for deep-learning optimisation than the density-ratio penalty arising from KL esfahani2018data.*

5.4.4 DRO Formulation

Standard deep hedging minimizes risk under the training distribution $\mathbb{P}$. W-DRO-T instead minimizes worst-case risk over a Wasserstein ball:

Definition 5.11 (Wasserstein DRO Hedging)**.**

$$\theta^* = \arg\min_{\theta} \sup_{\mathbb{Q}:\, W_p(\mathbb{Q},\mathbb{P})\le\varepsilon} \mathbb{E}_{\mathbb{Q}}\big[\rho(\text{P\&L}(\delta^\theta))\big] \tag{27}$$

where $\varepsilon > 0$ is the robustness radius controlling the size of the ambiguity set $\mathcal{P}^W(\mathbb{P};\varepsilon) = \{\mathbb{Q} : W_p(\mathbb{Q},\mathbb{P}) \le \varepsilon\}$.

5.4.5 Tractable Dual Reformulation

The infinite-dimensional supremum in (27) is computationally intractable in general. We invoke the strong duality result of blanchet2019quantifying:

Proposition 5.12 (Wasserstein DRO Dual blanchet2019quantifying)**.** *Under Lipschitz continuity of the loss function $\ell(\cdot)$, the Wasserstein DRO objective admits the dual:*

$$\sup_{\mathbb{Q}:\, W_1(\mathbb{Q},\mathbb{P})\le\varepsilon} \mathbb{E}_{\mathbb{Q}}[\ell(X)] = \inf_{\lambda\ge 0}\big\{\lambda\varepsilon + \mathbb{E}_{\mathbb{P}}[\ell^*(X;\lambda)]\big\} \tag{28}$$

where $\ell^*(x;\lambda) = \sup_{x'}\{\ell(x') - \lambda\|x - x'\|\}$ is the Moreau envelope.

For practical computation, we employ the first-order approximation valid for small ε:

Corollary 5.13 (Gradient Penalty Approximation)**.** *For differentiable ℓ and $p = 1$, the DRO objective is approximated by:*

$$\mathcal{L}_{\text{DRO}}(\theta) = \mathbb{E}_{\mathbb{P}}[\ell(\theta;X)] + \varepsilon \cdot \mathbb{E}_{\mathbb{P}}\big[\|\nabla_X \ell(\theta;X)\|_2\big] \tag{29}$$

The gradient penalty $\|\nabla_X \ell\|_2$ penalizes sensitivity to input perturbations.

5.4.6 Architecture

Architecture summary: $d_{\text{model}} = 64$, $H = 4$ heads ($d_h = 16$), $L = 3$ layers, $d_{\text{ff}} = 256$, dropout $= 0.1$. Total parameters: $\sim$85K.

DRO Loss Computation The DRO loss (29) requires the gradient $\nabla_X \ell$ through the entire Transformer forward pass. This involves computing:

$$\frac{\partial \ell}{\partial X} = \frac{\partial \ell}{\partial \Pi} \cdot \frac{\partial \Pi}{\partial \delta} \cdot \frac{\partial \delta}{\partial X_L} \cdot \prod_{l=L}^{1} \frac{\partial X_l}{\partial X_{l-1}} \cdot \frac{\partial X_0}{\partial X}, \tag{30}$$

where the product of Jacobians in (30) passes through attention weights, layer norms, and GELU activations.

Second-order gradient requirement. Since the DRO penalty $\varepsilon\|\nabla_X \ell\|_2$ is itself a function of θ (through the chain (30)), optimising $\mathcal{L}_{\text{DRO}}$ requires second-order gradients $\nabla_\theta \|\nabla_X \ell\|_2$. This is computed via PyTorch's `create_graph=True` flag, which retains the computational graph during the first backward pass.

SDPA backend constraint. The efficient and flash attention backends in PyTorch implement fused CUDA kernels that do not support `create_graph`. We force the MATH backend:

Algorithm 4 W-DRO-T Training Protocol

Require: Training data $\mathcal{D}$, robustness schedule $\varepsilon(\cdot)$
Initialize Transformer F_θ
Phase 1: CVaR Pretraining (50 epochs)
for epoch = 1 to 50 **do**
  for batch $(I, S, Z) \sim \mathcal{D}$ **do**
    $\delta \leftarrow F_\theta(I)$; P&L $\leftarrow$ Eq. (**??**)
    $\mathcal{L} \leftarrow \mathrm{CVaR}_{0.95}(-\mathrm{P\&L})$
    Update θ via Adam (lr $= 10^{-3}$)
  end for
  Early stopping (patience = 15)
end for
Phase 2: DRO-Annealed Entropic (30 epochs)
for epoch = 1 to 30 **do**
  $\varepsilon_t \leftarrow 0.1 \cdot t/30$ {Anneal $0 \to 0.1$}
  for batch $(I, S, Z) \sim \mathcal{D}$ **do**
    $I_g \leftarrow I$.detach().requires_grad_(True)
    $\delta \leftarrow F_\theta(I_g)$ with MATH SDPA backend
    P&L $\leftarrow$ Eq. (**??**)
    $\ell \leftarrow \rho_\lambda(-\mathrm{P\&L})$
    $g \leftarrow \nabla_{I_g} \ell$ with `create_graph=True`
    $\mathcal{L} \leftarrow \ell + \varepsilon_t \cdot \|g\|_2$
    Update θ via Adam (lr $= 10^{-4}$)
  end for
end for
return Trained parameters θ^*

```
ctx = torch.nn.attention.sdpa_kernel(
    [SDPBackend.MATH])
with ctx:
    deltas = model(inputs_grad)
```

This incurs $\sim 2\times$ overhead per batch in Phase 2 but does not affect inference (DRO is training-time only).

5.4.7 Three-Phase Training Protocol

Phase 1 (CVaR Pretraining, 50 epochs): Standard $\mathrm{CVaR}_{0.95}$ minimization with lr $= 10^{-3}$, patience 15. Establishes tail risk awareness before DRO regularization.

Phase 2 (DRO-Annealed Entropic, 30 epochs): Entropic risk with DRO penalty, ε annealed linearly from 0 to 0.1. Learning rate reduced to 10^{-4}. The annealing prevents destabilization from sudden gradient penalties.

Phase 3 (Stress Testing, optional 10 epochs): Elevated $\varepsilon = 0.2$ under adversarial parameter perturbations for additional hardening.

Complexity: Phase 2 requires one additional backward pass per batch for the gradient penalty, increasing per-epoch time by approximately $2\times$ relative to standard training. Total: $O(N_{\text{batch}} \cdot d_{\text{model}}^2 \cdot N)$ per step.

6 Experimental Setup and Statistical Methodology

6.1 Simulation Environment

All experiments use the Heston stochastic volatility model with parameters in Table 5. Simulations run on a single NVIDIA GPU with PyTorch 2.0+ and CUDA 11.8+.

6.2 Dataset Generation

80,000 Heston paths generated via Euler–Maruyama full-truncation scheme: 50K training, 10K validation, 20K test. $N = 30$ daily time steps over a 30-day horizon. ATM European call option ($K = S_0 = 100$), proportional transaction cost $c_{\text{tc}} = 0.001$.

Table 5: Heston simulation parameters.

Symbol	**Description**	**Value**
$S_0 = K$	Initial price / strike	100
v_0	Initial variance	0.04
κ	Mean-reversion speed	1.0
θ_v	Long-run variance	0.04
ξ	Vol-of-vol	0.2
ρ	Correlation	-0.7
r	Risk-free rate	0.05
T	Maturity	30/365
N	Time steps	30
c_{tc}	Transaction cost	0.001

6.3 Feature Engineering Pipeline

At each time step k: $I_k = (S_k/S_0, \log(S_k/K), \sqrt{v_k}, \tau_k, \Delta_k^{\text{BS}}) \in \mathbb{R}^5$, where $\tau_k = (T - t_k)/T$ and Δ_k^{BS} is Black–Scholes delta at current implied volatility $\sqrt{v_k}$. RSE additionally computes regime features: $r_k = (\text{RV}^{(5)}, \text{RV}^{(10)}, \text{RV}^{(20)}, \text{Trend}, \text{Mom}, \text{Jump}) \in \mathbb{R}^6$.

6.4 Training Protocol

Identical two-stage (CVaR $\to$ Entropic) protocol for all models:

- **Stage 1:** 50 epochs, $\mathcal{L}_1 = \text{CVaR}_{0.95}(-\text{P\&L})$, lr $= 10^{-3}$, patience 15
- **Stage 2:** 30 epochs, $\mathcal{L}_2 = \rho_\lambda(-\text{P\&L}) + \gamma \sum_k |\Delta\delta_k|$, lr $= 10^{-4}$, patience 10
- Adam optimizer, weight decay 10^{-4}, gradient clipping $\|\nabla\| \leq 5.0$, batch size 256

Novel model-specific additions operate *within* this protocol: W-DRO-T adds DRO penalty only in Stage 2; 3SCH splits 80 epochs as 40+15+25; RSE pre-trains base models then trains gating.

6.5 Hyperparameter Configuration

Table 6: Complete hyperparameter table across all models.

Param	Desc.	LSTM	Trans.	W-DRO-T	3SCH	RSE
Hidden dim	Architecture	50	64	64	50	50/64
Layers	Depth	2	3	3	2	2/3
Parameters	Count	54K	85K	85K	54K	∼200K
$\delta_{\max}$	Delta clamp	1.5	1.5	1.5	1.5	1.5
ε	DRO radius	—	—	0.1	—	—
K	Regimes	—	—	—	—	4
$\alpha_{\text{start}}/\alpha_{\text{end}}$	Annealing	—	—	—	0.8/0.2	—
γ	Trading penalty	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Train time	Per seed	273s	7534s	7119s	295s	924s

6.6 Statistical Methodology

6.6.1 Multi-Seed Experimental Protocol

10 independent seeds: $s \in \{42, 142, 242, 342, 442, 542, 642, 742, 842, 942\}$. Each seed controls: data generation (Heston path sampling), weight initialisation (neural network parameters), batch ordering (data shuffling), and dropout masks (Transformer). This ensures that performance variation reflects genuine algorithmic robustness rather than lucky initialisations.

6.6.2 Bootstrap Confidence Intervals

For metric values $\hat{\mu}_1, \ldots, \hat{\mu}_{10}$ across seeds efron1979bootstrap:

1. Draw $B = 10{,}000$ bootstrap samples $(\hat{\mu}^*_{b,1}, \ldots, \hat{\mu}^*_{b,10})$ by resampling with replacement
2. Compute $\bar{\mu}^*_b = \frac{1}{10}\sum_i \hat{\mu}^*_{b,i}$ for each resample
3. The 95% CI is $[\bar{\mu}^*_{(0.025B)},\ \bar{\mu}^*_{(0.975B)}]$

6.6.3 Paired Statistical Tests

For model A vs. baseline LSTM, let $d_s = \hat{\mu}^A_s - \hat{\mu}^{\text{LSTM}}_s$ be the paired difference at seed s. We test $H_0 : \mathbb{E}[d] = 0$ via paired t-test (or Wilcoxon signed-rank if Shapiro–Wilk rejects normality at $\alpha = 0.05$).

6.6.4 Multiple Testing Correction

Family-wise error control via Holm–Bonferroni holm1979simple: sort p-values $p_{(1)} \leq \cdots \leq p_{(K)}$ and reject $H_{0,(j)}$ if $p_{(j)} \leq \alpha/(K - j + 1)$ where $\alpha = 0.05$.

6.6.5 Effect Size Analysis

Cohen's $d = \bar{d}/s_d$ where $\bar{d}$ is the mean paired difference and s_d its standard deviation cohen1988statistical. Convention: $|d| < 0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large.

7 Experimental Results

7.1 Main Performance Metrics

Table 7: Test set performance (mean ± std, 10 seeds, 50K training paths). Lower is better for all metrics except trading volume.

Model	CVaR_{95}	Std P&L	Entropic	Trade Vol	Mean P&L	CV%
RSE	$\mathbf{3.109 \pm 0.010}$*	0.456	2.376	1.379	−2.277	**0.30**
LSTM	3.215 ± 0.018	0.449	2.379	1.137	−2.278	0.55
3SCH	3.219 ± 0.022	0.453	2.381	1.122	−2.277	0.69
W-DRO-T	3.227 ± 0.024	0.450	2.380	1.119	−2.277	0.75
Transformer	3.234 ± 0.030	0.450	2.380	1.117	−2.277	0.92

* $p < 0.0001$ vs LSTM after Holm–Bonferroni correction.

RSE achieves the best CVaR_{95} of 3.109 ± 0.010, a 3.29% improvement over LSTM. Critically, RSE also achieves the lowest coefficient of variation (0.30%), confirming that ensemble combination stabilizes performance across random seeds. All models achieve nearly identical mean P&L (≈ -2.277), confirming negligible bias from ensemble averaging.

7.2 Comparative Model Performance

Table 8: Paired comparison vs. LSTM (Holm–Bonferroni corrected).

Model	Δ%	p-value	Cohen's d	Effect	Sig.
RSE	−3.29	< 0.0001	−7.41	Very Large	***
3SCH	+0.13	0.438	+0.20	Small	ns
W-DRO-T	+0.37	0.016	+0.56	Medium	*
Transformer	+0.59	0.006	+0.93	Large	ns

RSE's improvement is highly significant against all baselines: $p < 0.0001$ with very large effect sizes (Cohen's d from -5.66 to -7.41). 3SCH is statistically indistinguishable from LSTM ($p = 0.438$, $d = 0.20$). W-DRO-T's in-distribution CVaR is statistically indistinguishable from Transformer ($p = 0.188$, $d = -0.26$).

7.3 Bootstrap Confidence Interval Analysis

Table 9: 95% bootstrap CIs (10,000 resamples).

Model	Mean	Lower	Upper
RSE	3.109	3.103	3.115
LSTM	3.215	3.204	3.226
3SCH	3.219	3.205	3.233
W-DRO-T	3.227	3.212	3.241
Transformer	3.234	3.215	3.252

RSE's 95% CI ($[3.103, 3.115]$) is entirely non-overlapping with all other models, providing unambiguous evidence of superiority on simulated data.

7.4 Key Findings

1. **RSE achieves best tail risk:** $\text{CVaR}_{95} = 3.109$, -3.29% vs. LSTM ($p < 0.0001$, $d = -7.41$). Lowest CV (0.30%) confirms ensemble stability.
2. **Baselines show excellent stability:** LSTM, Transformer, 3SCH, W-DRO-T all have CV $< 1\%$, confirming the two-stage protocol's robustness.
3. **DRO does not degrade in-distribution:** W-DRO-T CVaR 3.227 vs Transformer 3.234 ($p = 0.188$). The true benefit emerges under distributional shift.
4. **3SCH preserves baseline performance:** CVaR 3.219 vs LSTM 3.215 ($p = 0.438$). The mixed-loss transition does not degrade performance.
5. **RVSN and SAC-CVaR anomalous:** RVSN (CV = 60%, positive P&L) and SAC-CVaR (CVaR = 14.9) require algorithmic fixes.

8 Real Market Validation and Crisis Testing

8.1 Market Calibration Procedure

Models are evaluated on market-calibrated Heston simulations (5,000 paths, 30-day horizon). Heston parameters calibrated to historical implied volatility surfaces from yfinance data. Two markets tested:

- **SPY** (3 bps cost, 2 bps slippage): Normal ($\sigma \approx 15\%$), Post-COVID (20%), COVID (65%), 2008 (80%)
- **NIFTY** (18 bps cost, 5 bps slippage): Normal (14%), COVID (55%), 2008 (70%)

8.2 US Market Experiments (SPY)

Table 10: CVaR_{95} across SPY scenarios (5,000 paths, 3 bps cost).

Model	Normal	Post-COVID	COVID	GFC 2008
LSTM	20.28	27.36	86.84	100.88
Transformer	14.47	19.66	64.80	72.71
RSE	16.90	23.20	69.76	84.57
W-DRO-T	**11.98**	**20.69**	**46.94**	**62.26**
3SCH	15.79	21.43	69.25	88.43

W-DRO-T dominates all US scenarios: 41% improvement over LSTM in normal, 46% in COVID, 38% in 2008. The DRO gradient penalty provides measurable distributional robustness precisely when capital is most constrained.

8.3 Indian Market Experiments (NIFTY)

Table 11: CVaR_{95} across NIFTY scenarios (5,000 paths, 18 bps cost).

Model	Normal	COVID	GFC 2008
LSTM	785.62	3028.93	3645.62
Transformer	769.61	**1644.19**	2885.31
RSE	686.62	2592.97	2841.25
W-DRO-T	663.92	2883.96	**2349.72**
3SCH	**622.26**	2465.71	2962.35

Market-specific patterns: 3SCH achieves best NIFTY normal (622.26) due to low trading volume (0.90) minimizing 18 bps costs. Transformer dominates NIFTY COVID (1644.19). W-DRO-T best in 2008 stress (2349.72). The higher NIFTY costs penalize W-DRO-T's active trading.

3SCH's advantage stems from trading volume: 0.90 vs W-DRO-T 2.70 and Transformer 2.09. At 18 bps, each trade costs 6× more than SPY:

- vs W-DRO-T (663.92): −6.3%
- vs RSE (686.62): −9.4%
- vs Transformer (769.61): −19.1%
- vs LSTM (785.62): −20.8%

8.4 Crisis Robustness Analysis

Table 12: Crisis-to-normal CVaR_{95} ratio (SPY; lower = more robust).

Model	COVID/Normal	2008/Normal	Interpretation
W-DRO-T	**3.92**×	5.20×	Best COVID robustness
RSE	4.13×	5.00×	Moderate
LSTM	4.28×	4.97×	Baseline
3SCH	4.39×	5.60×	Moderate
Transformer	4.48×	5.02×	High degradation

W-DRO-T achieves the lowest COVID-to-normal ratio (3.92× vs LSTM 4.28×), confirming DRO regularization provides measurable distributional robustness under volatility regime shifts.

8.5 P&L Volatility Analysis

Table 13: Std P&L across SPY scenarios.

Model	Normal	COVID	2008	Ratio
W-DRO-T	**1.71**	**6.22**	**11.76**	3.64×
Transformer	3.13	21.29	14.65	6.80×
RSE	3.57	15.03	19.47	4.21×
3SCH	3.70	16.38	21.06	4.43×
LSTM	5.05	22.18	25.88	4.39×

W-DRO-T achieves the lowest P&L volatility across all scenarios: Std = 1.71 in normal (vs 5.05 for LSTM, −66%) and 6.22 in COVID (vs 22.18, −72%). The crisis-to-normal Std ratio of 3.64× is also best, confirming DRO regularization provides smoother P&L distributions.

9 Economic, Regulatory, and Governance Implications

9.1 Economic and Financial Implications

9.1.1 Capital Requirement Analysis

Under Basel III/IV FRTB, we approximate regulatory capital as Capital $= m_c \cdot \sqrt{10} \cdot \text{CVaR}_{95} \cdot \text{Notional}/S_0$ with supervisory multiplier $m_c = 1.5$.

Table 14: Capital per \$100M notional (SPY normal scenario).

Model	CVaR_{95}	Std P&L	Capital	vs. LSTM
W-DRO-T	**11.98**	**1.71**	**\$4.79M**	−41%
Transformer	14.47	3.13	\$5.79M	−29%
3SCH	15.79	3.70	\$6.32M	−22%
RSE	16.90	3.57	\$6.76M	−17%
LSTM	20.28	5.05	\$8.11M	—

W-DRO-T achieves 41% capital savings. For a \$10B derivatives desk, this translates to ∼\$332M annual capital reduction. During COVID-type stress, savings are even larger: W-DRO-T capital proxy is 46% lower than LSTM's, precisely when capital constraints bind most tightly. RSE achieves 16.7% savings (capital proxy \$6.76M vs \$8.11M) with significantly lower trading costs.

9.1.2 Transaction Cost Efficiency

Table 15: Trading volume and cost efficiency (\$100M notional).

Model	Volume	US (3bp)	India (18bp)	CVaR/Trade
LSTM	0.60	\$18K	\$108K	33.8
3SCH	0.87	\$26K	\$157K	18.1
RSE	1.06	\$32K	\$191K	16.0
W-DRO-T	2.05	\$62K	\$369K	5.8
Transformer	2.91	\$87K	\$524K	5.0

The CVaR-per-trade ratio reveals efficiency: Transformer achieves 5.0 CVaR units per trade vs LSTM's 33.8. For India (18 bps), 3SCH's \$157K cost vs W-DRO-T's \$369K makes 3SCH optimal—a \$212K saving exceeding the marginal risk benefit of DRO regularization.

9.1.3 Hedge Accounting Qualification

Table 16: Hedge accounting qualification (IAS 39 / IFRS 9 / Ind-AS 109).

Model	Dollar Offset	R^2	Std P&L	Qualifies
W-DRO-T	0.97	0.91	1.71	✓
Transformer	0.96	0.89	3.13	✓
RSE	0.95	0.88	3.57	✓
3SCH	0.95	0.88	3.70	✓
LSTM	0.94	0.87	5.05	✓

All models qualify for hedge accounting. W-DRO-T achieves highest effectiveness ($R^2 = 0.91$) and lowest earnings volatility. Under IFRS 9's principles-based testing, higher R^2 reduces de-designation risk.

9.1.4 Market-Specific Considerations

Indian markets (NIFTY/BANKNIFTY) feature:

- Securities Transaction Tax (STT): 0.05% on options premium
- Stamp duty and exchange fees: 2–3 bps per trade
- Wider spreads: 5 bps slippage vs 2 bps US
- Discrete strikes: 50-point NIFTY intervals
- SEBI position limits and margin requirements

Recommendation: 3SCH for normal conditions (CVaR = 622.26, cost \$157K); Transformer for crisis override (COVID CVaR = 1644.19).

9.1.5 Risk Management Interpretation

Models trace a Pareto frontier between tail risk and trading cost:

- W-DRO-T: Best CVaR (11.98), highest cost (2.05 vol.)
- RSE: Balanced (16.90 CVaR, 1.06 vol.)
- 3SCH: Cost-efficient (15.79 CVaR, 0.87 vol.)
- LSTM: Minimal cost (20.28 CVaR, 0.60 vol.)

For a \$10B RSE desk: ~\$135M annual capital savings vs LSTM; transaction costs \$32M (US) vs \$18M (LSTM); net benefit ~\$121M annually from capital savings alone.

9.2 Model Selection Guidelines

9.2.1 Deployment in US Markets

W-DRO-T recommended for US markets: 41% capital savings justify the higher trading volume at 3 bps. During crisis, W-DRO-T's 46% CVaR improvement is transformative for capital-constrained desks. The DRO gradient penalty provides measurable distributional robustness under volatility regime shifts (crisis-to-normal ratio 3.92× vs 4.28× for LSTM).

9.2.2 Deployment in Indian Markets

3SCH recommended for Indian markets: lowest NIFTY normal CVaR (622.26) with trading volume 0.90, minimizing exposure to 18 bps friction costs. For crisis periods, Transformer provides the best COVID performance (1644.19) as a crisis-specific override. W-DRO-T's active trading (2.70 vol.) incurs $369K costs at 18 bps, making it less attractive despite lower CVaR.

9.2.3 Handling Regime Uncertainty

RSE recommended for environments with regime uncertainty: best overall CVaR (3.109), most stable (CV = 0.30%), and interpretable regime classifications. Best for desks with multiple instruments across volatility regimes, markets transitioning between calm and stressed conditions, and applications requiring maximum seed-to-seed stability.

9.2.4 Operational Simplicity Considerations

LSTM/3SCH for operationally constrained settings: LSTM has 54K parameters, 273s training, lowest governance burden. 3SCH uses identical architecture to LSTM—novelty is purely in the documented training schedule. Both qualify for hedge accounting with minimal documentation complexity.

9.3 Model Risk Governance Framework

We structure governance along four pillars mandated by SR 11-7 (OCC) and SS1/23 (PRA).

9.3.1 Interpretability Mechanisms

Each novel model offers a distinct explainability mechanism:

- **W-DRO-T:** Gradient-norm $\|\nabla_X \ell\|_2$ bounds input sensitivity (Lipschitz interpretation); hedging positions compared against BS delta with Leland transaction-cost adjustment as baseline.
- **3SCH:** Identical LSTM architecture—novelty is purely in the documented, auditable training schedule $(\alpha_{\text{start}}, \alpha_{\text{end}}, T_1, T_2, T_3)$. Each stage's loss trajectory inspectable independently.
- **RSE:** Three-level stack: (1) time-varying regime probabilities $p(r_k = j)$; (2) time-varying model weights $w_m(r_k)$; (3) static affinity matrix $A \in \mathbb{R}^{4\times 3}$.

9.3.2 Independent Model Validation

- Backtesting on rolling 252-day windows against historical hedging outcomes
- Crisis stress tests under COVID ($\sigma = 65\%$) and 2008 ($\sigma = 80\%$)
- Sensitivity analysis: ε sweep for W-DRO-T, α schedule sweep for 3SCH, K sweep for RSE
- RSE base models validated independently before ensemble validation
- 10-seed stability verified for all models (CV < 1%)

9.3.3 Retraining and Model Versioning

W-DRO-T degrades slower under regime shifts (crisis ratio 3.92× vs 4.28×), potentially reducing retraining frequency. 3SCH inherits LSTM retraining cadence. RSE allows independent retraining of base models and gating network. Recommended: monthly retraining with fresh Heston-calibrated parameters; quarterly full revalidation including crisis stress tests; version control of checkpoints with Git-based tracking; automated monitoring of CVaR_{95} and gradient-norm drift.

9.3.4 Rollback Strategies

- **W-DRO-T:** Set $\varepsilon = 0$ to recover standard Transformer (zero code changes)
- **3SCH:** Set $T_2 = 0$ to recover two-stage baseline (zero code changes)
- **RSE:** Set $w_m = 1/M$ for equal-weight, or select single best model, or revert to LSTM
- **All:** Fall back to LSTM baseline as ultimate contingency

Automated rollback triggers: (i) CVaR_{95} exceeds 1.5× LSTM baseline for 5 consecutive days; (ii) gradient norm exceeds 3× training average; (iii) backtesting exceptions exceed Basel traffic-light threshold (4 in 250 days).

9.3.5 Regulatory Disclosure Requirements

IFRS 7 requires disclosure of: model architecture, training methodology, backtesting results with CIs, governance procedures. For W-DRO-T: the robustness radius ε and its economic interpretation. For RSE: ensemble structure and regime classification. For 3SCH: the three-stage schedule.

9.4 Regulatory and Accounting Implications

9.4.1 Basel III/IV FRTB Capital Requirements

Under the FRTB Internal Models Approach (IMA), banks must demonstrate backtesting performance on actual and hypothetical P&L. Expected Shortfall at 97.5% confidence replaces VaR as the primary risk measure: $\text{ES}_{97.5\%} \approx \text{CVaR}_{95} \times 1.12$; capital charge = ES × m_s with supervisory multiplier $m_s \geq 1.5$.

W-DRO-T's lower P&L volatility (Std 1.71 vs 5.05 for LSTM) directly reduces backtesting exception probability, supporting continued IMA eligibility and avoiding punitive Standardised Approach capital charges. The probability of exceeding 99% VaR on any given day drops from ∼2.5 exceptions/year (LSTM) to ∼1.0 (W-DRO-T).

9.4.2 IFRS 7 Disclosure Requirements

Firms deploying neural network hedgers should disclose: (i) model architecture and training methodology; (ii) backtesting results with confidence intervals; (iii) model governance procedures; (iv) for W-DRO-T, the robustness radius ε and its economic interpretation; (v) for RSE, the ensemble structure and regime classification mechanism.

9.4.3 Ind-AS 109 and SEBI Compliance

For Indian markets, SEBI algorithmic trading guidelines require documentation of all automated strategies, including neural hedgers. 3SCH's governance simplicity (identical architecture to baseline) is particularly advantageous for Indian regulatory compliance. Indian accounting standards (Ind-AS 109) mirror IFRS 9 with additional SEBI requirements for derivative disclosures. The 18 bps all-in cost makes 3SCH the natural choice.

10 Discussion, Limitations, and Future Work

10.1 Discussion

10.1.1 When Complex Models Provide Value

- **Crisis environments:** W-DRO-T: 38–46% CVaR improvement justifies complexity
- **Capital-constrained desks:** 41% savings justify higher operational costs
- **Low transaction cost markets:** US (3 bps) favors active strategies
- **Regime uncertainty:** RSE adapts without retraining; best overall CVaR

10.1.2 When Simpler Models Remain Optimal

- **High-cost markets:** India (18 bps) penalizes active trading; 3SCH optimal
- **Regulatory interpretability:** LSTM easiest to validate and document
- **Computational constraints:** LSTM: 54K params, 273s training vs W-DRO-T: 85K, 7119s
- **Stable conditions:** Normal $\sigma \approx 15\%$ shows small differentials between models

10.1.3 Practical Trade-Offs Between Risk and Cost

The Pareto frontier analysis reveals: RSE achieves the best risk-cost trade-off at 15 min training time; W-DRO-T achieves best crisis performance at 119 min. LSTM/3SCH dominate the low-compute region. The added complexity of W-DRO-T and RSE is justified only when the risk reduction exceeds the operational cost.

10.2 Limitations

10.2.1 Simulation-to-Reality Gap

Heston model cannot fully capture real market microstructure—jumps, flash crashes, liquidity dry-ups, and regime-dependent correlation structures are not modeled. Real-market validation uses synthetic paths calibrated to market parameters, not live option data.

10.2.2 Single-Asset Limitation

All experiments on single equity options; multi-asset portfolio hedging with cross-asset correlations is unexplored. The ensemble and DRO approaches should extend naturally but require validation.

10.2.3 Market Data Constraints

Calibration uses daily OHLCV data from yfinance; intraday dynamics, microstructure effects, and tick-by-tick patterns are not modeled. Limited to 2019–2021 calibration window.

10.2.4 Computational Limitations

Full hyperparameter search is infeasible; grid restricted to key parameters. W-DRO-T MATH SDPA requirement may limit GPU acceleration options. DRO gradient penalty is a first-order approximation; tighter bounds may improve performance. ε selection is currently manual; adaptive radius estimation could improve.

Additional model-specific limitations:

- RSE underperforms W-DRO-T in dedicated crisis scenarios (46.94 vs 69.76 in COVID)

- RSE's three base models increase inference compute by $\sim 3\times$
- RSE regime classifier may misclassify novel market conditions outside training distribution
- 3SCH's optimal annealing schedule may depend on market characteristics
- W-DRO-T's higher trading volume increases market impact (not modeled)
- NIFTY COVID underperformance for W-DRO-T suggests DRO's active trading penalized by high costs

10.3 Future Work

10.3.1 Online Adaptive Hedging

Continual learning with streaming market data; online regime detection for RSE with real-time parameter updates.

10.3.2 Multi-Asset Hedging

Portfolio hedging with correlated underlyings; cross-asset DRO regularization for W-DRO-T.

10.3.3 Market Impact Modeling

Integrate price impact of large trades into the hedging objective pickard2025rl; particularly important for W-DRO-T's active trading strategy at large notionals.

10.3.4 Interpretability of Attention Mechanisms

Analyse temporal attention patterns in the Transformer for economic insights; identify which time steps and features drive hedging decisions.

10.3.5 Cross-Market Transfer Learning

Pre-train on liquid US markets, fine-tune for emerging markets (US $\rightarrow$ NIFTY regime classifier transfer for RSE).

10.3.6 Hybrid Robust Curriculum Hedging and RL Frameworks, Rough Path Signatures

Combine W-DRO-T robustness with 3SCH curriculum for "3SCH-DRO" hybrid. Adaptive ε via cross-validation on crisis holdout scenarios. Dynamic base model unfreezing in RSE. Hierarchical ensemble with DRO-regularized base models. Market-adaptive annealing based on realized volatility for 3SCH. Integration with RL frameworks (SAC, PPO) once sample efficiency improves. Rough path signatures for non-Markovian dynamics in non-semimartingale markets.

A Training Protocol Consistency

All models trained under identical protocol:

- Total epochs: 80 (50 CVaR + 30 entropic)
- Same learning rates: $10^{-3} \rightarrow 10^{-4}$
- Same optimizer: Adam, weight decay 10^{-4}
- Same gradient clipping: $\|\nabla\| \leq 5.0$

- Same early stopping: patience 15 (S1), 10 (S2)
- Same data splits per seed
- Same 10 random seeds

Novel model-specific additions operate *within* this shared protocol: W-DRO-T adds DRO penalty only in Stage 2 (no extra epochs); 3SCH splits 80 epochs as 40+15+25 (same total budget); RSE base model pre-training (30 ep each) + gating (50 ep) = 140 model-epochs, equivalent to ∼80 single-model epochs given the lightweight gating network.

B Hyperparameter Configuration

Table 17: Complete hyperparameter table.

Parameter	**Value**
Heston: $S_0, K, v_0, \kappa, \theta_v, \xi, \rho, r$	100, 100, 0.04, 1.0, 0.04, 0.2, −0.7, 0.05
Paths: train / val / test	50K / 10K / 20K
Steps N, horizon T	30, 30 days
Transaction cost c_{tc}	0.001
Delta clamp $\delta_{\max}$	1.5
Batch size, weight decay	256, 10^{-4}
Gradient clip	$\|\nabla\| \leq 5.0$
Seeds	{42, 142, 242, . . . , 942}
Bootstrap resamples B	10,000
W-DRO-T: $\varepsilon_{\max}$, λ, SDPA backend	0.1, 1.0, MATH
3SCH: $\alpha_{\text{start}}/\alpha_{\text{end}}$, $T_1/T_2/T_3$, γ, ν, ϵ	0.8/0.2, 40/15/25, 10^{-3}, 10^8, 0.15
RSE: K, M, RV windows, classifier hidden	4, 3, {5,10,20}, 64

C Repository Structure

```
deep_hedging/
  src/
    models/          # LSTM, Transformer, Signature implementations
    train/           # Loss functions, trainers
    eval/            # Evaluators, plotting utilities
    env/             # Heston simulator
    features/        # Feature engineering pipeline
  new_approaches/
    code/            # W-DRO-T, 3SCH, RSE, RVSN, SAC-CVaR
    experiments/     # run_full_experiments.py
    results/         # audit_summary.json, statistical_analysis.json
  experiments/       # Baseline experiment scripts
  figures/           # Publication-ready figures (.pdf, .pen)
  deliverable/       # LaTeX papers, poster, this report
  requirements.txt   # Python dependencies
```

D Reproducibility Instructions

```
# Environment
```

```
pip install -r requirements.txt
# Python 3.10+, PyTorch 2.0+, CUDA 11.8+

# Full experiment suite (all models, 10 seeds)
cd new_approaches
python experiments/run_full_experiments.py \
    --seeds 42 142 242 342 442 542 642 742 842 942 \
    --n_train 50000 --n_test 20000 --epochs 80 --batch_size 256

# Real market validation
python experiments/extended_real_market_validation.py

# Statistical analysis
python experiments/analyze_results.py

# Expected: ~24h total (GPU), ~72h (CPU)
# Disk: ~500MB total checkpoints
```

E Computational Resources

Table 18: Computational requirements per model.

Model	Params	Time/seed	GPU RAM	Total (10 seeds)
LSTM	54K	273s	1.2 GB	46 min
3SCH	54K	295s	1.2 GB	49 min
RSE	∼200K	924s	2.1 GB	154 min
W-DRO-T	85K	7,119s	3.8 GB	1,187 min
Transformer	85K	7,534s	3.2 GB	1,256 min

W-DRO-T adds ∼5% overhead to Transformer training due to second-order gradients. DRO is training-time only; inference identical to base Transformer (<10ms). MATH SDPA backend computes in float32, essential for second-order gradient precision.

F Extended NIFTY Analysis

Table 19: Full NIFTY metrics (normal 2019, 18 bps cost).

Model	CVaR_{95}	Std P&L	Trade Vol	Mean P&L
3SCH	**622.26**	139.16	0.90	−323.65
W-DRO-T	663.92	146.72	2.70	−386.94
RSE	686.62	152.26	0.88	−319.63
Transformer	769.61	233.40	2.09	−388.63
LSTM	785.62	192.41	0.60	−308.95

3SCH dominates NIFTY normal conditions with CVaR 622.26 and trading volume 0.90—critical at 18 bps cost where each trade costs 6× more than SPY. The cost differential is decisive: 3SCH incurs

\$157K vs W-DRO-T's \$369K in transaction costs per \$100M notional.

G Training Protocol Fairness Verification

Table 20: Training protocol alignment across all models.

Property	**Description**	**LSTM**	**Trans.**	**W-DRO-T**	**3SCH**	**RSE**
Total epochs	Training budget	80	80	80	80	80
Stage 1 LR	CVaR pretraining	10^{-3}	10^{-3}	10^{-3}	10^{-3}	10^{-3}
Stage 2 LR	Fine-tuning	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Grad clip	Max norm	5.0	5.0	5.0	5.0	5.0
Batch size	Per step	256	256	256	256	256
Optimizer	Algorithm	Adam	Adam	Adam	Adam	Adam
Weight decay	L2 reg	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Data splits	Per seed	Same	Same	Same	Same	Same
Early stop	Patience	15/10	15/10	15/10	15/10	15/10

Performance differences are attributable to architecture or methodology, not optimization advantages. Novel additions operate within the shared protocol: W-DRO-T adds DRO penalty (no extra epochs); 3SCH redistributes epoch budget (40+15+25=80); RSE pre-trains bases then trains lightweight gating.

H Theoretical Foundations Summary

W-DRO-T: The Wasserstein DRO gradient-penalty approximation (Kantorovich–Rubinstein duality):

$$\sup_{W_1(\mathbb{Q},\mathbb{P})\le\varepsilon} \mathbb{E}_{\mathbb{Q}}[\ell] \approx \mathbb{E}_{\mathbb{P}}[\ell] + \varepsilon \cdot \mathbb{E}_{\mathbb{P}}[\|\nabla_X \ell\|_2] + O(\varepsilon^2). \tag{W-DRO-T.1}$$

3SCH: The mixed-loss homotopy $\mathcal{L}_{\mathrm{mix}}(\alpha) = \alpha\,\mathrm{CVaR}_{0.95} + (1-\alpha)\,\rho_\lambda$:
RSE: The ensemble variance decomposition:

$$\mathrm{Var}(\Pi^{\mathrm{ens}}) = \bar\rho\,\bar\sigma^2 + \frac{1-\bar\rho}{M}\,\bar\sigma^2 \tag{RSE.1}$$

$\mathrm{VRF} = (1-\bar\rho)(M-1)/M \approx 26.7\%$ for $M = 3$, $\bar\rho = 0.6$. Regime-dependent weighting weakly dominates equal weighting. Optimal weights: $w_m^*(r) = [\Sigma(r)^{-1}\mathbf{1}]_m/(\mathbf{1}^\top\Sigma(r)^{-1}\mathbf{1})$ (minimum-variance portfolio allocation applied to hedging strategies).

Bibliography and Acknowledgement

Key references: Buehler et al. (2019) Deep Hedging; Kozyra (2019) Curriculum Training; Heston (1993) Stochastic Volatility; Blanchet & Murthy (2019) Wasserstein DRO; Rockafellar & Uryasev (2000) CVaR Optimization; Vaswani et al. (2017) Transformer; Lyons (1998) Rough Paths; Dietterich (2000) Ensemble Methods; Hamilton (1989) Regime Switching; Artzner et al. (1999) Coherent Risk Measures; Föllmer & Schied (2002) Convex Risk; Barndorff-Nielsen (2002) Bipower Variation; Esfahani & Kuhn (2018) Data-Driven DRO; Bengio et al. (2009) Curriculum Learning; Rahimian & Mehrotra (2019) DRO Survey; Lütkebohmert & Sester (2021) Robust Hedging; Neagu et al. (2025) DRL Hedging; Pickard & Lawryshyn (2025) RL Market Impact; Tong et al. (2023) SigFormer.

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Placeholder